



Data Extraction and Preprocessing for Automated Question Answering Based on Knowledge Graphs

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Abstract. This article discusses the algorithms and methods of data representation for knowledge graphs. The proposed algorithms make it possible to automate the process of extracting and processing data from users' requests within the mixed learning process and reduce the role of an expert in the preparation of question-answering data sets necessary for training models of dialogue systems. The results show that the method of enrichment of the knowledge graph leads to an increase in the number of links and the accuracy of vector representation models.

Keywords: Semantic technology · Ontology · RDF · Embeddings · QA system

1 Introduction

Dialog systems that are a technological trend have recently received growing attention. Nowadays, the most popular language technology is an intelligent virtual assistant. Depending on the application context, such systems are called a dialog interface, question-answering system, or chatbot. However, they all share the same concept because they are made to perform specific tasks by way of interactive communication with machines and queries in a natural language [1]. Among the most famous dialog systems are digital voice assistants, Apple Siri, Amazon Alexa, Google Assistant, Yandex Alice. Their emergence stimulated the growth of many text-based dialog systems designed to solve concrete problems. It can be argued that dialog systems have become the de facto standard for many industries, from retail to booking, and, in particular, customer support.

One of the areas of dialog system application is education, especially the development of e-learning and distance learning technologies in the implementation of educational programs. The Federal Law on Education in the Russian Federation describes possible ways of integration and organization educational process with the use of distance learning technologies and e-learning based on Unified Educational Information Environments (UEIE).

Development of such projects and their direct integration in educational processes of the leading Russian universities leads to a significant increase in both

the number of educational materials and the number of students, which in turn generates large amounts of data. It also increases the number of support staff for such systems, including curators of online courses and support specialists competent in a specific field or subject. An increase in the number of employees involved in this process entails economic costs and efficiency evaluations.

Generalization of UEIE information, including study materials, tasks, tests, exercises, user messages on forums, support requests in the form of one knowledge base that relies on a semantic network, can solve the efficiency problem. Unlike databases available in any UEIE, knowledge bases not only store factual information, but also make conclusions on existing or newly introduced facts and, thereby, perform semantic or meaningful information processing.

1.1 Motivation

The knowledgebase can be a base for the development of an intelligent system, one example of which is the creation of a question answering system. Their development is a goal of many corporations, in particular, in banking. Potentially, dialog systems will save money by automating repetitive tasks while supporting various processes of user interaction with the system.

In education, it is the communication between a student and a teacher on the educational forum or by e-mail. However, these questions are often repeated. Thus, they can be formalized to automate the answering process. The second important advantage is the ability to provide feedback 24 hours a day, unlike support staff. Conversational systems also allow gaining experience by analyzing conversations.

The current problem of interactive systems is the closed nature of many developments and commercialization. In particular, the important problem is the preprocessing of data for interactive system models to learn from.

Therefore, the problem is to generalize and develop methods and algorithms that will ensure a continuous process of developing a working model of the dialog system using raw, semi-structured data of support resources.

1.2 Dialog Systems

Recent advances in dialogue systems are overwhelmingly driven by deep learning techniques that have been used to improve a wide range of big data applications, such as computer vision, natural language processing, and recommendation systems. For conversational systems, deep learning can use a huge amount of data to learn meaningful function representations and response generation strategies, while requiring a minimum amount of manual processing.

Goal-oriented dialog systems based on the **pipeline method** typically consist of the following four components [2]:

1. Natural Language Understanding, NLU. It splits user text into predefined semantic slots and converts it to a machine-readable form.

2. Dialog State Tracker, DST. It defines the current state, which is used to select the next action.
3. Dialog Policy Learning, DPL. It selects the next action based on the current state of the dialog.
4. Natural Language Generation, NLG. It displays the selected action due to the generated natural language response.

As a rule, non-target dialogue systems do not try to solve a specific problem and give an answer to a question, but their initial goal is to maintain a dialogue with a person on an arbitrary topic. The approaches to constructing **non-target** dialog systems, including the following:

1. **Generative approach** that is based on neural networks. This approach requires large amounts of data for training, while the question answering pairs must be fully correct. The advantage of the approach is the ability to generate answers close to the natural language in cases when the answer is not contained in the data set.
2. The **search approach** is based on the principle that the answer is inside the dataset, and the model goal is to find the most suitable one.
3. The **ensemble approach (hybrid)** involves combining methods to improve the responses generated by the neural network through providing an additional context by using the options offered by the search approach.

1.3 Knowledge Graphs

The approach based on knowledge graphs is actively developing and becoming extremely popular among modern dialog systems [4]. A knowledge graph is a set of structured information ordered in a graph format, where nodes are entities and edges are relationships. The knowledge graph $\mathcal{G}$ consists of triples of the form ‘subject–predicate–object’ – tuples (h, r, t) , where $h(head)$ – subject, $r(relation)$ – predicate, $t(tail)$ – object. Such triples are called true triples and they form a set $\mathcal{S} = \{(h, r, t)\}$. Herewith $h, t \in \mathcal{E}$ and $r \in \mathcal{R}$, where $\mathcal{E}$ – set of entities, and $\mathcal{R}$ – set of relations.

To create a system based on knowledge graphs, it is necessary to present data in the appropriate triples format, as well as to process user requests from proposals, in the triples format, for example, from a question: ‘In what year was St. Petersburg founded?’ triple can be obtained ‘St. Petersburg – founded – year’. Such a triple is then compared with the data available in the knowledge graph and place of the year property, values will be substituted.

Thus, knowledge graphs allow us to model both abstract logical statements and schemes, and to fill these schemes with particular objects of the real world [13]. Moreover, knowledge graphs allow machines to reason and derive new knowledge not previously described in the graph. Formal semi-structuring and powerful logical apparatus distinguish knowledge graphs from traditional relational databases, which are structured, that is, have established relationships and relationships.

2 Method

The primary use case of developing and maintaining methods and algorithms for the automated processing of semi-structured support resources is to automate the preparation of a training set for the dialog system. The described processing is based on natural language processing and machine learning methods, which allows selecting the text from e-mails with the subsequent extraction of question answering pairs.

2.1 Data Preprocessing

The basic algorithm is collecting and preprocessing data. Automatically collected emails require format conversion. Initially, these messages are transmitted in the Multipurpose Internet Mail Extension (MIME) format. The basic format for email messages is defined by the RFC822 standard. According to it, each message consists of head and content. The head stores service information, namely the address necessary to send the message, while the messages can be stored in any format, and the standard only determines the format for transmission.

Messages are collected in the base64 byte-string format. It is a sequence of printed ASCII characters (Latin characters, numbers, and special characters). Thus, each email must be decoded in the UTF-8 format because it is the most common and convenient data encoding format. Next, the message text is read line by line and cleared of extraneous data. It includes HTML and CSS markup based on tags.

The main task of the subsequent processing is to separate the user's signature from the message because its availability and format are not regulated by representation standards in any way. For this, a series of heuristics have been developed to classify message lines:

1. **Regular Expressions** is a simple but effective way [12] to find standardized information: web addresses, telephones, markers of the message signature department, email address, full name.
2. **String selection** is a criterion for the length of message signature lines, determined on the basis of marked up messages.
3. **Keywords** are a criterion based on the assumption that a certain set of words is used in the message signature.

Thus, to identify keywords, the received messages are divided into tokens, each of which is compared with a list of stop words, and then reduced to normal form. As a result, all documents (email) are presented as a bag of words that are further analyzed.

Based on the obtained heuristics the problem of binary classification is solved. The problem can be solved by common methods of machine learning with a different number of processed last k lines of the message. The best result is obtained with $k = 10$.

2.2 Question Extraction

The algorithm for extracting question answering pairs based on the following criteria:

- the sentence ends with a question mark;
- the sentence contains a word or phrase with a semantic load defined as a pair of a noun and a verb by way of extracting universal dependencies [5] (an example of the corresponding representation is shown in the Fig. 1);
- the sentence contains pronominal adverbs.

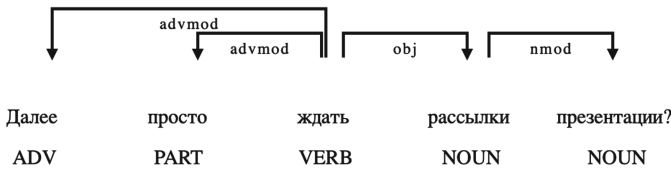


Fig. 1. Representation of a sentence using universal dependencies for Russian. The translation into English: Should I wait for the email with a presentation?

These criteria are combined:

- last symbols e ‘?’ and e contains a noun and a verb;
- first three or more characters e contained in $\mathcal{ADV}_q$ and e contains a noun and a verb.

The algorithm produces identifiers of messages and questions. Message identifiers are used for searching the respective answers. Such information is contained in the RFC822 transmission format of messages.

2.3 Knowledge Graph Enrichment

Based on the obtained data we introduce the process of creating a basic ontological model for a knowledge graph of support resources, taking into account its metadata enrichment, topic modeling, named entities recognition (NER), and semantically related questions.

The sequence of actions for applying topic modeling methods includes tokenization and stemming or lemmatization of requests. As a result, each request is presented as a bag of words $\mathcal{D}$. To ensure the accuracy of the model, words with a frequency of more than one are selected to cut off common words, and focus the topic model within a specific context.

In the next step, we define the determination of the sentence semantic proximity based on word proximity. To define it, the Russian language thesaurus RuWordNet [7] based on the RuThes thesaurus [6] that was converted to WordNet [8] was used. The thesaurus contains 111500 words and expressions

in Russian in total. To mark parts of speech, an approach based on Universal Dependencies [5] was used.

The semantic vector can be obtained by way of the pairwise comparison of the words included in the sentence with each other, where each coordinate of the vector corresponds to the maximum value of the proximity of pairwise words [9]. The obtained value is normalized based on the criterion of the synonymy threshold that is proposed to be equal to 0.8025 in the paper [10].

For all question pairs in the knowledge graph $\mathcal{G}$, the algorithm produces the added triplets of the form $(\mathcal{S}_1, \text{SimilarToTheQuestion}, \mathcal{S}_2)$ with a set threshold for proximity of sentences γ . This parameter can be established empirically only and must take high values to ensure that only true identical questions are close.

NER extraction allows us to enrich the knowledge graph with metadata specific to certain types of questions. It is proposed to use the BERT model [11] available for Russian and having the best results at the time of the experiment. Identified entities are added to the knowledge graph $\mathcal{G}$, and are associated with an email and/or question depending on the location of the named entity. One entity can be associated with any questions or email.

Finally, we are using the TransE [3] vector representation algorithm to prove the relevance and necessity of the knowledge graph and its enrichment with the specified metadata. The main idea of the algorithm is to represent the triples (h, r, t) of the set $\mathcal{S}$ in a certain vector space of dimension R^k , where k is the hyperparameter of the model. All entities h, t form the set $\mathcal{E}$, all relationships r form the set $\mathcal{R}$. Embeddings are usually designated by the corresponding bold characters $\mathbf{h}, \mathbf{r}, \mathbf{t}$. Then, for the true triples, the condition $\mathbf{h} + \mathbf{r} \approx \mathbf{t}$ must be satisfied while being in contradiction with the false triples. The set of false triples $\mathcal{S}'$ is formed by substitution of h or t (but not both) of the true triplet for a random entity, that is:

$$\mathcal{S}' = \{(h', r, t) | h' \in \mathcal{E}\} \cup \{(h, r, t') | t' \in \mathcal{E}\}.$$

The search for such vector coordinates is performed using gradient descent while minimizing the loss function

$$\mathcal{L} = \sum_{(h, r, t) \in \mathcal{S}} \sum_{(h', r, t') \in \mathcal{S}'_{(h, r, t)}} [\gamma + d(\mathbf{h} + \mathbf{r}, \mathbf{t}) - d(\mathbf{h}' + \mathbf{r}, \mathbf{t}')]_+,$$

where d is a measure of similarity, $[x]_+$ means that only positive terms are considered, $\gamma > 0$ is an indent hyperparameter.

3 Results

The modeling was performed using four machine learning methods, and the results were evaluated using a confusion matrix presented on the Fig. 2.

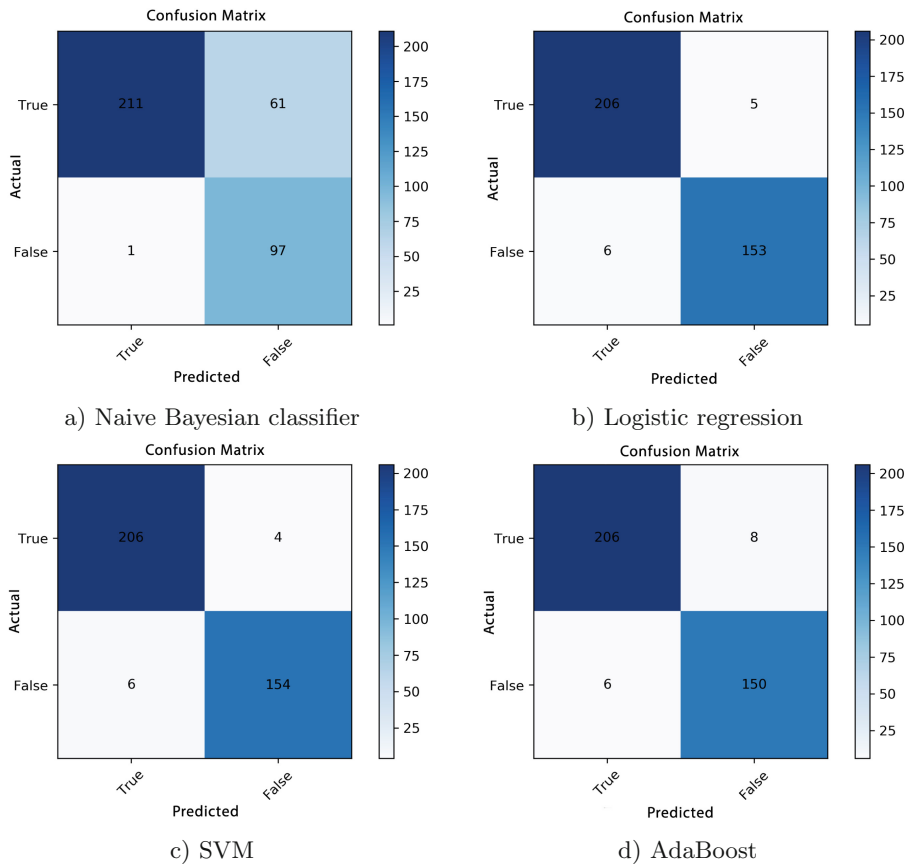


Fig. 2. Confusion matrix for the signature string classifier, $k = 10$.

The best result is achieved when $k = 10$ using the support vector machine (SVM) $F1 = 0.9763$. The Fig. 2 shows the confusion matrices for the selected classification methods based on model verification on a test data set from 370 marked up messages (212 messages contain a signature, 158 do not contain a signature). At the same time, the results of the basic algorithm adapted for Russian (translated keywords and regular expressions adapted for the Cyrillic alphabet) with the same parameters shows the worst result $F1 = 0.7332$.

Thus, the processing sequence of the developed algorithm allows receiving the processed support requests. Compared to the basic algorithm, the obtained metrics allow estimating the proposed approach effectiveness. The accuracy increased from $F1 = 0.7332$ to $F1 = 0.9763$. The increase was about 33.16%.

Table 1. Metrics for the developed algorithm, $k = 10$.

Algorithm	Metrics		
	F1	Precision	Recall
Naive Bayesian Classifier	0.8719	0.7757	0.9953
Logistic Regression	0.9740	0.9763	0.9717
SVM	0.9763	0.9810	0.9717
AdaBoost	0.9671	0.9626	0.9717

Based on described methods we perform a series of experiments, based on the different states of the knowledge graph (Table 1):

- The basic structure of the knowledge graph – $\mathcal{G}_1$;
- The knowledge graph $\mathcal{G}_1$ enriched with named entities – $\mathcal{G}_2$;
- The knowledge graph $\mathcal{G}_2$ enriched with semantic proximity of questions – $\mathcal{G}_3$;
- The knowledge graph $\mathcal{G}_3$ enriched with the topic of requests as a result of topic modeling – $\mathcal{G}_4$;
- The knowledge graph $\mathcal{G}_4$ enriched with metadata of requests – $\mathcal{G}$.

The Table 2 shows the results of the model accuracy evaluation.

Table 2. Model accuracy for different knowledge graphs.

Method	Metrics	
	F1	EM
DeepPavlov [14]	0.2341	0.1172
$\mathcal{G}_1$	0.2763(+18.03%)	0.1493(+27.42%)
$\mathcal{G}_2$	0.2764(+0.04%)	0.1493(+0.00%)
$\mathcal{G}_3, \gamma = 0.73$	0.2902(+4.00%)	0.1531(+2.53%)
$\mathcal{G}_4$	0.4179(+44.00%)	0.2136(+39.51%)
$\mathcal{G}$	0.5057(+21.01%)	0.2250(+5.31%)

4 Conclusions

Thus, the problems stated in the article were solved making it possible to prove the effectiveness of the developed algorithms and method in the experimental study based on the accuracy assessment criteria.

The proposed methods were compared based on various embedding approaches and concerning basic experiment without knowledge graphs. The proposed methods showed higher accuracy results. The lack of direct analogs or methods that can be used to develop dialog systems in Russian for the automated processing of resource data emphasizes the relevance of the paper.

Therefore, the goal of improving the accuracy of answers in dialog systems during automated processing of resource data by the means of the semantic annotation method and metadata enrichment as well as embedding algorithms of knowledge graphs was achieved.

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